

Three Ways to Judge an Agent-Native Research Artifact

A metascience comparison of GRO ideal metrics, an ARA-inferred metric design (v3), and a semantic verifier (Seal Level 2) over 12 compiled ARAs

0. Abstract

We compare three distinct approaches to evaluating **Agent-Native Research Artifacts (ARAs)** — machine-readable knowledge packages compiled from published papers — over a fixed corpus of **12 ARAs** in the Alzheimer's disease / neurodegeneration domain. The three approaches occupy different **rigor tiers**:

1. **GRO ideal metrics (Tier A, deterministic + reliable-anchored + reproducible-judged)**. A battery of metrics computed over the ARA's typed graph (claims, quantities, references, entities, genre), plus a smaller set of externally anchored checks (live PubMed / registry lookups) run on the biomarker/trial half of the corpus. 2. **The first ARA-inferred metric design (v3)**. An earlier tournament-selected battery — `extractive_fidelity`, `six_paper_rankers`, and a composite `artifact_trust` — that layers a partial source-text semantic check on top of structural completeness logic. 3. **The ARA verifier (Seal Level 2)**. A calibrated semantic reviewer that reads the argument and scores six epistemic dimensions (D1–D6), producing a per-ARA recommendation from Strong Accept to Reject.

Three findings organize the paper. **First**, the approaches differ enormously in *reach*: GRO scores all 12/12 ARAs; v3 fully scores only 3/12; Seal scores all 12 but through human-grade semantic labor. **Second**, they measure *near-orthogonal constructs*: Seal's overall score is essentially uncorrelated with GRO's structural health (Spearman $\rho = -0.011$) and with v3 artifact-trust ($\rho = 0.035$), while the two structural instruments correlate strongly with each other ($\rho = 0.678$, $p = 0.015$). **Third**, no single approach discriminates good science from bad science on its own — each catches real defects the other two are structurally blind to. We argue the three tiers are **complementary, not competing**: a deterministic structural floor, an externally anchored middle, and a judged semantic ceiling. We are explicit about the limits: the anchored/judged evidence exists for only half the corpus, there is **no external good-vs-bad ground-truth set**, and $n = 12$ in a single sub-domain.

1. Setup

1.1 The 12-ARA corpus

All 12 ARAs were compiled from published Alzheimer's / neurodegeneration papers and carry GRO "ideal" data shapes (a typed claim/quantity/reference/entity/genre graph). They fall into two natural clusters that matter for everything downstream:

- A **biomarker + trial cluster** (6 ARAs): `che26`, `val25`, `han26`, `zim25`, `jes26`, `tit26` — the p-tau217 assay and donanemab-trial papers. **These are the only ARAs with any externally anchored or judged-tier evidence.**

- A **biology / epidemiology / pipeline cluster** (6 ARAs): gau25, zho25, aki26, cum26, xu25, ard25 — assessed on deterministic-tier metrics only.

Slug	Short description	Genre (GRO paper-type)	# claims
che26	Diagnostic performance of plasma p-tau217 (network meta-analysis)	systematic_review_meta_analysis	8
val25	Blood biomarkers of AD (SNAC-K observational cohort)	observational_epidemiology	13
han26	VeraBIND Tau — plasma tau pathological activity	diagnostic_accuracy_study	12
zim25	Donanemab in early symptomatic AD (long-term extension)	randomized_controlled_trial (LTE)	12
jes26	Efficacy and safety of donanemab	randomized_controlled_trial	10
tit26	Automated high-throughput plasma p-tau quantification	diagnostic_accuracy_study	11
gau25	Single-nucleus + spatial transcriptomic profiling	primary_experimental	14
zho25	Microglia networks (narrative review)	narrative_review_survey	14
aki26	Molecular signatures of resilience to AD	molecular_neuropathology_single_cell_omics	10
cum26	AD drug-development pipeline census	clinical_pipeline_registry_census	10
xu25	Epidemiological & sociodemographic transitions	epidemiological_secondary_analysis	13
ard25	Response of spatially defined microglia states	preclinical_mechanistic_study	13

1.2 The three approaches

- **GRO ideal metrics.** Eight deterministic metrics computed over the typed ARA graph, every one flagged `computable_on_prose: false` — they *only* exist because the compiler typed the graph, and they *never read the source PDF*. Plus four externally anchored/judged metrics (retractions, external reference-resolvability, prior-literature novelty, trial-registry concordance) run against live PubMed / ClinicalTrials.gov for the 6-ARA biomarker cluster.
- **v3 (first ARA-inferred design).** `extractive_fidelity` (an LLM re-reads the source and flags unsupported/contradicted/distorted claims), six `paper_rankers`, and a composite `artifact_trust` (`grounding_trust`, `environment_completeness`, `falsifiability_presence`, `compiler_fidelity`, `gate_pass_ratio`). v3 **degrades to pending rather than fabricating a score** when it lacks the input to judge.
- **Seal Level 2.** A semantic reviewer scoring six dimensions on a 1–5 scale — **D1 Evidence Relevance, D2 Falsifiability Quality, D3 Scope Calibration, D4 Argument Coherence, D5 Exploration Integrity, D6 Methodological Rigor** — then an overall score and a recommendation.

1.3 The three rigor tiers

The cleanest way to read the whole comparison is as three tiers of epistemic warrant:

Tier	What it answers	How	Primary instrument here
Tier A — deterministic	"Is the artifact internally well-formed?"	Pure lookup / arithmetic over typed fields; no comprehension, no external call	GRO deterministic metrics; v3 structural components
Tier B — reliable-anchored	"Do its external referents actually exist?"	Live query to an authority the author can't edit (PubMed, registry)	GRO retractions / external ref-resolution / registry
Tier C — reproducible-judged / semantic	"Is the underlying science any good?"	LLM/human reads the argument and judges	GRO novelty verdict (judged); Seal L2 (semantic)

The central empirical claim of this paper is that these tiers are **near-orthogonal in practice** and each is **necessary but not sufficient**.

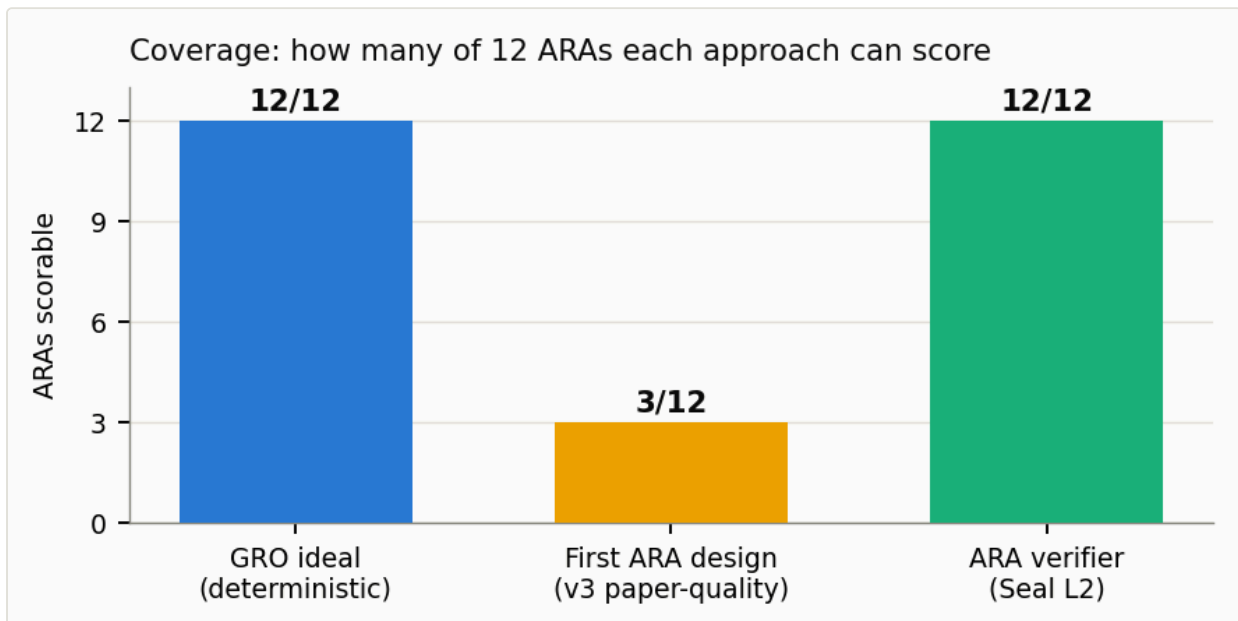


Figure 1. Coverage — how many of the 12 ARAs each approach can score.

2. The GRO ideal metrics: best vs. most uncertain

GRO's own analysis ranks its metrics by reliability. The top-3 are metrics that fire consistently and are hard to game; the bottom-3 are metrics whose *construct validity* is in doubt — including one that is directly falsified by GRO's own anchored check.

2.1 The three best

- claim_typing_coverage (deterministic).** 1.0 across all 12/12 ARAs, 0 nulls. A pure structural presence/enum check (`claim_type ∈ enum AND logical_form ≠ not_specified`) with no external dependency. Gaming it would require fabricating a non-empty logical form for *every* claim — a visible, checkable artifact. Zero variance across five very different genres is evidence the check is well-specified, not trivially true. **Caveat: this same zero variance means it has no discriminating power in this corpus.**
- quantity_reconciliation_rate (deterministic).** 1.0 across all 10/12 ARAs where it is defined (null for zho25, aki26 — structurally vacuous, no quantity cited by ≥ 2 claims). An arithmetic cross-check between two independently authored fields (the ledger's canonical value vs. the numeral embedded in the claim's own quote). The only blemish is coverage, not correctness.
- retractions (reliable-anchored).** 0 found across all 6 externally-checked

ARAs and their top-cited references, each verified via itemized live PubMed publication-type queries. Anchored to an authority the ARA author cannot edit. **Caveat: floor effect — the corpus contains no known-bad papers, so this is proven trustworthy, not proven discriminating.**

2.2 The three most uncertain

1. **reference_resolvability_rate (self-declared / deterministic).** The one metric with a direct ground-truth check on the *same quantity* — and the ground truth contradicts it. Declared corpus mean **0.5197** vs. independently observed **0.9583** (a ≈ 0.44 gap). val25 declares 0.0 (0/55) but PubMed resolved 9/12 sampled refs; han26 declares 0.19 but resolves 1.0; tit26 declares 0.132 but resolves 1.0. Per GRO's own conclusion, the deterministic metric "scores DOI-capture completeness rather than reference verifiability" — it measures whether the compiler captured a DOI at compile time, not whether the reference is real. 2. **Prior-literature novelty verdict (reproducible-judged).** 22.6% of checked claims (7/31) land in `cannot_determine`; verdicts split almost evenly across all four categories (novel 32.3% / incremental 32.3% / cannot_determine 22.6% / previously_reported 12.9%). Each verdict rests on an LLM judgment over a PubMed search, and the methodology surfaces its own limit — "absence of a PubMed hit is not strong evidence of absence." Only 31 claims across 6/12 ARAs are covered at all. 3. **Trial-registry concordance (anchored + judged).** Only 2 trial papers exist to check (zim25, jes26), and **both are post-hoc analyses of the exact same trial, NCT04437511** — so, per the source file, "registry-concordance evidence is effectively a single data point." The concordance-gap verdict (post-hoc CDR-Global HR, plasma p-tau217, LTE-vs-ADNI comparisons added beyond registered outcomes) is real but generalizes to nothing. 10/12 ARAs have registry = null.

2.3 Also flagged (construct-validity concerns)

- **genre_silent_omissions** has a documented formula bug: `count(expected) - count(present) - count(absent_declared)` can go **negative** (zho25 = -1) when a genre's `present_slots` includes a slot outside its own `expected_slots` list.
- **broken_ref_integrity** is 100% concentrated in one ARA (ard25 = 16/16); ard25's own diagnostics attribute this to a missing `logic/experiments.md` file, not genuinely broken citations — the metric conflates "cites a nonexistent id" with "cites a real id in a ledger the compiler never found."
- **overclaim_flags** concentrates almost entirely in two genres (xu25 epidemiological projections = 11/13; cum26 pipeline census = 9/10) that structurally traffic in population-level estimates lacking a single sample-size field — raising the question of whether it flags overclaiming or just a genre feature.
- **entity_anchoring_rate** = 0.0 for all 12/12 ARAs — a corpus-wide pipeline gap (no ARA does ontology linking), reliably measured but with zero discriminating power.

2.4 Per-metric summary

GRO metric	Tier	Corpus stat	Coverage	Reliability verdict
claim_typing_coverage	deterministic	mean 1.0 (min 1.0, max 1.0)	12/12	Best — but non-discriminating
quantity_reconciliation_rate	deterministic	mean 1.0	10/12 (2 null)	Best — coverage gap only
retractions	anchored	0 found	6/12	Best — floor effect
entailment_precondition_rate	deterministic	mean 0.821 (0.4–1.0)	12/12	Middle — real spread, useful
external reference-resolvability	anchored	observed mean 0.958	6/12	Middle — sound check, small sample
genre_silent_omissions	deterministic	total 4 (min –1, max 4)	3/12 nonzero	Middle — formula defect (can go negative)
overclaim_flags	deterministic*	total 23 (max 11)	3/12 nonzero	Middle — genre-blindness / construct doubt
broken_ref_integrity	deterministic	total 16 (all ard25)	1/12 nonzero	Middle — single-ARA compilation artifact
entity_anchoring_rate	deterministic	0.0 everywhere	12/12	Middle — reliable, zero discrimination
reference_resolvability_rate	deterministic	mean 0.374 (0.0–1.0)	12/12	Most uncertain — falsified by anchored check
prior-lit novelty verdict	judged	31 claims, 22.6% cannot_determine	6/12	Most uncertain — search-phrasing dependent
trial-registry concordance	anchored + judged	2 papers, same trial	2/12	Most uncertain — n = 1 trial

*overclaim_flags is nominally deterministic but scored over LLM-typed population_scope / population_n fields.

2.5 The full deterministic grid

Every deterministic metric, every ARA. CTC and EAR are constant (1.0 and 0.0) and carry no discriminating information; the action is in the middle columns.

ARA	QuantRecon	Entailment	RefResolv	GenreOmit	Overclaim	BrokenRef
che26	1.00	1.00	0.944	0	0	0
val25	1.00	0.692	0.000	1	0	0
han26	1.00	1.00	0.190	0	0	0
zim25	1.00	1.00	0.957	0	0	0
jes26	1.00	1.00	0.895	0	0	0
tit26	1.00	1.00	0.132	0	0	0
gau25	1.00	0.571	0.100	0	0	0
zho25	null	0.500	0.000	–1	3	0
aki26	null	0.400	0.136	0	0	0
cum26	1.00	1.00	1.000	0	9	0
xu25	1.00	0.923	0.137	0	11	0
ard25	1.00	0.769	0.000	4	0	16

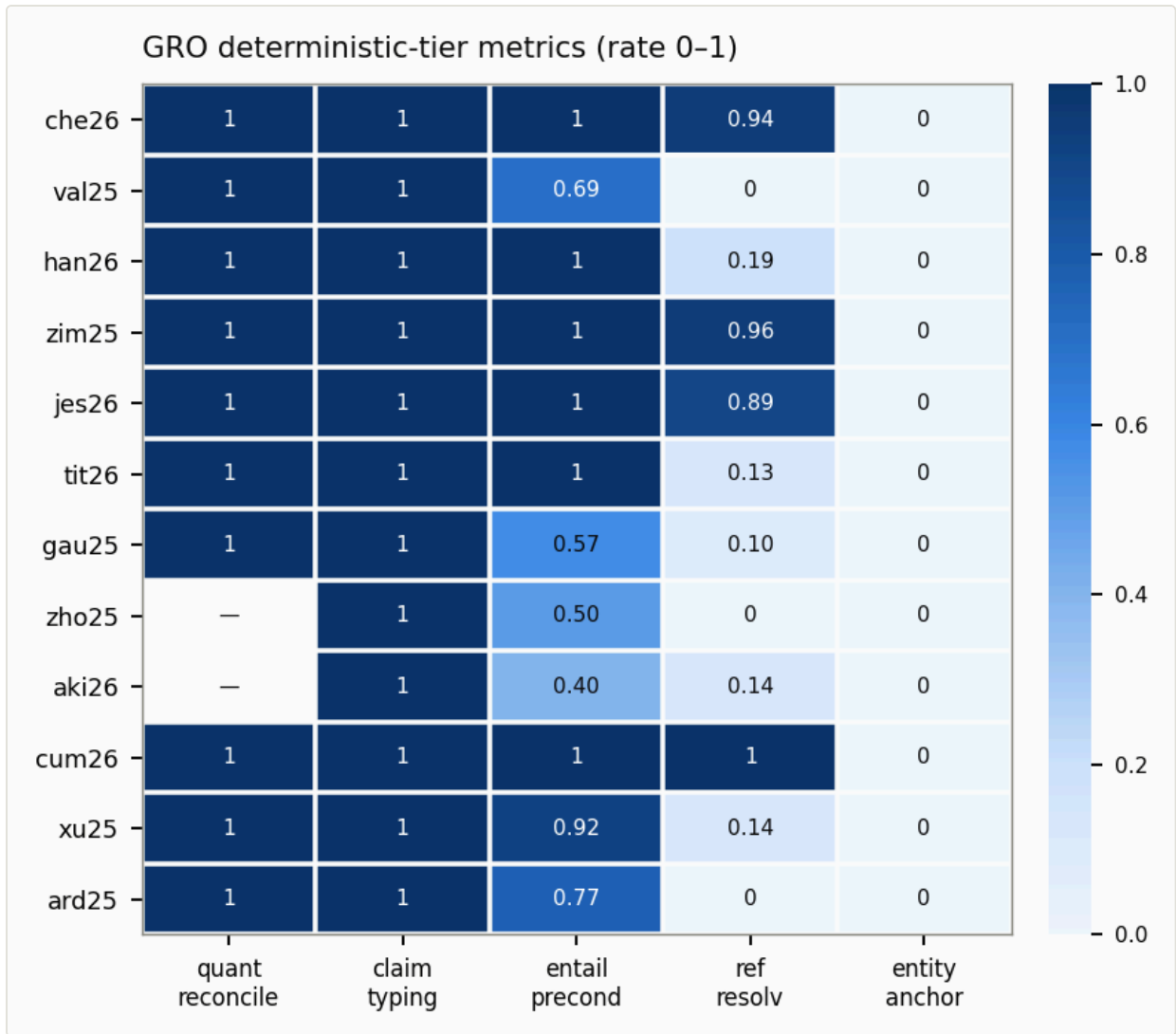


Figure 2. GRO deterministic-tier metrics per ARA (rate 0-1; “—” = undefined).

3. Held against the first ARA-inferred design (v3)

3.1 The coverage gap is architectural

v3 could **fully score 3/12 ARAs (25%)**; GRO scored **all 12/12 (100%)**.

- v3 `rank_eligible = true`: only che26, tit26, cum26.
- v3 `extractive_fidelity` computed (non-pending): 4/12 — che26 (0.625), jes26 (0.9), tit26 (1.0), cum26 (1.0); pending on the other 8/12.
- v3 `semantic_grounding` judge-scored: 4/12; pending on 8/12.
- GRO: 12/12 processed, 0 failed; the only metric-level nulls are `quantity_reconciliation_rate` on zho25 and aki26 (structurally vacuous).

The gap is not incidental. **Every GRO metric is `computable_on_prose: false` and computes over the typed graph that exists on every compiled ARA.** v3's `extractive_fidelity` and `paper_rankers` additionally require an LLM to re-read the source PDF and judge fidelity/novelty — a pass wired up (frozen judge findings) for only 3–4 ARAs. So **v3 trades reach for depth** (it verifies extraction against source prose, but only where the pass was run) while **GRO trades depth**

for reach (it runs everywhere, but only checks the ARA's own internal consistency and never sees the source paper).

3.2 Construct overlap

Construct	v3 metric	GRO metric	Relationship
Claim-to-source fidelity	<code>extractive_fidelity</code> (re-reads source PDF)	<code>quantity_reconciliation_rate</code> + <code>entailment_precondition_rate</code> (ledger self-consistency only)	GRO strictly weaker — it cannot catch a claim that is faithful to its own record but was never true of the source. v3's is the only construct in either system that re-checks the source text.
Reference / citation integrity	<code>artifact_trust.gates.all_links_resolve</code> (boolean)	<code>reference_resolvability_rate</code> (continuous) + <code>broken_ref_integrity</code> (count)	Same target; GRO more diagnostic (continuous rate/count vs. a single pass/fail bit).
Claim scope / overreach	<code>semantic_grounding</code> (judge-scored, 4/12)	<code>overclaim_flags</code> (mechanical scope-label vs. missing-n)	Different axis of the same worry; GRO's runs everywhere but is a narrow, genre-correlated proxy.
Genre-appropriate completeness	genre used only as an eligibility label	<code>genre_silent_omissions</code>	GRO adds this construct outright; v3 has no analog.

GRO-only constructs (no v3 analog): `entity_anchoring_rate` (0.0 corpus-wide), `broken_ref_integrity`, `genre_silent_omissions`, `overclaim_flags`. **v3-only constructs (no GRO analog):** `corrective_science_score`, `dead_end_density` / `negative_result_share`, `translation_trial_linkage`, `novel_claim_count` — the paper rankers.

v3's own tournament headline — **0/6 usable paper rankers** — is confirmed on this corpus: `corrective_science_score` is `invalid_fabricated` on 12/12; `negative_result_share` is `pending_sem` on 12/12; `translation_trial_linkage` is self-annotated "superseded" on 12/12; `dead_end_density` is `fabricated` or `vacuous`; `semantic_grounding` is judge-scored on only 4/12; `novel_claim_count` is `pending_sem` even where a value exists. None reaches a fully valid corpus-wide state.

3.3 Agreement and divergence

- **Sharpest divergence — che26.** v3 `extractive_fidelity` = 0.625 catches a **fabricated forest-plot statistic** (MD = 0.10, 95% CI 0.04–0.16) that appears nowhere in the source. GRO scores this same ARA `quantity_reconciliation_rate` = 1.0 and `entailment_precondition_rate` = 1.0 — clean — because those metrics only check the ledger against its own quote and never touch the PDF. **GRO's structural checks cannot substitute for source verification even where both nominally target "fidelity."**
- **Second-sharpest divergence — cum26.** v3's *best* ARA (fidelity 1.0, `semantic_grounding` 0.941, 8/8 NCTs verified) is GRO's *most-overclaiming* ARA (`overclaim_flags` = 9/10). Perfectly faithful extraction of a source that itself makes unscoped population claims.
- **tit26.** v3 marks it clean (fidelity 1.0, all gates pass), but GRO's continuous `reference_resolvability_rate` = 0.132 exposes that most references are unverifiable — a distinction v3's boolean gate cannot make.

- **Sharpest agreement — ard25.** v3 gates (`seal_l1 = false`, `all_links_resolve = false`, `low_trust = true`) and GRO counts (`broken_ref_integrity = 16`, `genre_silent_omissions = 4`, both corpus maxima) converge on the same "structurally worst" verdict — GRO supplying the auditable counts v3's booleans compress away.

4. Held against the ARA verifier (Seal L2)

4.1 Per-ARA Seal scorecard

Seal dimensions: **D1** Evidence Relevance · **D2** Falsifiability Quality · **D3** Scope Calibration · **D4** Argument Coherence · **D5** Exploration Integrity · **D6** Methodological Rigor (each 1–5).

ARA	D1	D2	D3	D4	D5	D6	Overall	Recommendation	GRO health	v3 trust
che26	4	4	3	4	4	3	3.7	Weak Accept	0.982	0.944
val25	5	4	5	5	4	5	4.7	Strong Accept	0.564	0.615
han26	4	4	4	4	5	3	4.0	Weak Accept	0.730	1.000
zim25	4	3	4	5	4	2	3.7	Weak Accept	0.986	0.931
jes26	4	4	4	5	3	3	3.8	Weak Accept	0.965	0.974
tit26	4	4	4	5	3	4	4.0	Accept	0.711	0.917
gau25	4	4	5	5	4	4	4.3	Accept	0.557	0.685
zho25	4	3	4	4	2	2	3.2	Weak Accept	0.143	0.178
aki26	4	4	4	5	5	4	4.3	Accept	0.268	0.919
cum26	4	3	4	5	4	4	4.0	Accept	0.550	0.970
xu25	4	4	4	5	5	3	4.2	Accept	0.264	0.754
ard25	4	3	4	4	1	2	3.0	Reject	0.000	0.150

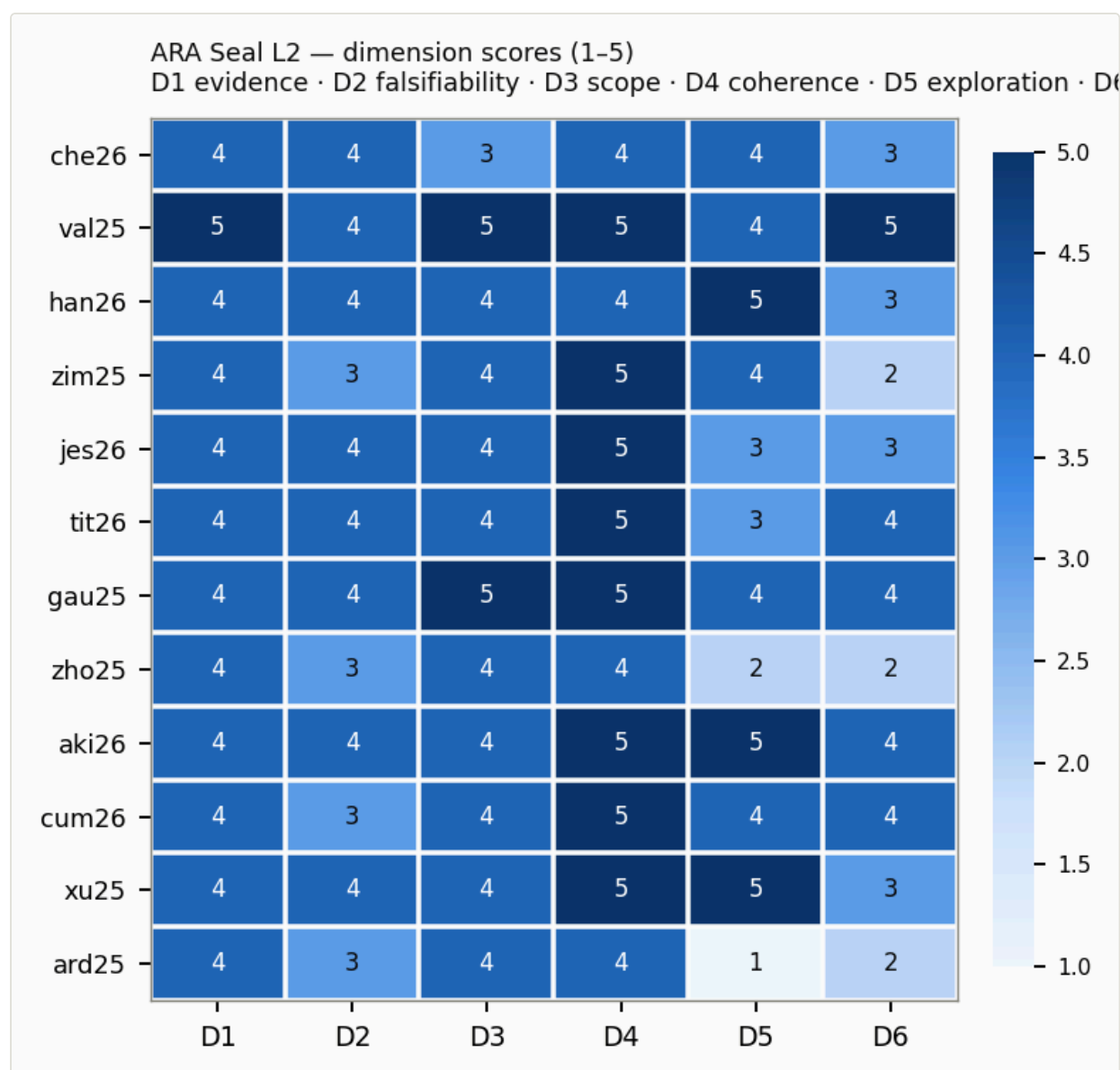


Figure 3. ARA Seal L2 dimension scores (1–5) per ARA.

4.2 What Seal catches that neither metric set can

- ****Scope over-generalization with a *stated* n (che26, D3 = 3).**** GRO `overclaim_flags` = 0 here, because its rule fires only when `population_n` is literally missing. Seal flags C05 verbatim: "*a generalization claim that overreaches its evidence type ... a P-score of 1.00 computed within only two Han Chinese cohorts (Huashan N=297, Greater Bay N=425) with no reported formal ethnicity-by-biomarker interaction test.*" The cohort sizes are named, so the structural rule is satisfied — detecting the overreach requires knowing what statistical test the claim needs. Comprehension, not a field lookup.
- **Evidence-source mismatch (che26).** Co3's "reduce the need for tau-PET scans by ~80%" is "*sourced from an unrelated external reference (Therriault et al., 2021) ... not derived from this NMA's own AUC/P-score outputs.*" No structural field encodes argument coherence.
- **Uncorrected multiplicity (jes26, zim25, val25).** A recurring methodological judgment: "*No multiplicity correction is applied ... yet every claim is nonetheless labeled 'supported' with uniform confidence.*" No GRO or v3 field maps to it.

- **Sponsor / conflict-of-interest and reproducibility (zim25, D6 = 2).** *"Sponsor-only authorship (all 18 authors are Eli Lilly employees) ... no released analysis code, no stated software version ... several supplementary figures/tables ... absent from the provided PDF."* No structural field maps to authorship or funding conflicts.

4.3 What the deterministic metrics catch that Seal misses or softens

- **Citation-graph integrity — no Seal dimension asks it.** val25 has **0/55 references resolvable** (`reference_resolvability_rate` = 0.0), yet Seal's D1 = 5 and overall = 4.7 (Strong Accept). Not a contradiction — D1 asks whether quoted evidence matches the claim, not whether the works-cited ledger resolves externally — but a reader of the Seal alone would never learn the bibliography doesn't resolve. Same blind spot: zho25 (0.0), ard25 (0.0), gau25 (0.1), tit26 (0.132).
- **Exact auditable counts where Seal narrates.** ard25: GRO `broken_ref_integrity` = 16/16 (100% of `proof_refs`). Seal converges narratively — *"every claim's verification-plan linkage is a dangling reference"* (D5 = 1) — but GRO supplies the precise fraction Seal's prose compresses into a floor-rule Reject.
- **Genre-blindness cutting the other way.** GRO `overclaim_flags` fires on 9/10 (cum26) and 11/13 (xu25) claims; Seal's D3 = 4 for both does **not** corroborate an overclaiming problem — these are census/GBD genres that legitimately state population-level estimates without a sample size. Here the *deterministic* metric is the one over-flagging.
- **A formula breaking at a genre edge.** zho25 `genre_silent_omissions` = -1 (a nonsensical negative from the literal count formula on a review genre); Seal instead reasons in prose that exploration integrity is *"honestly disclaimed rather than fabricated ... effectively unscorable"* (D5 = 2). Genre-aware judgment the rigid formula cannot make.

4.4 Do the three rank consistently?

Spearman rank correlation across the 12 ARAs:

Pair	ρ (all 12)	ρ	ρ (excl. ard25, n=11)
Seal overall vs GRO health	-0.011	0.974	-0.318
Seal overall vs v3 artifact-trust	0.035	0.913	-0.258
GRO health vs v3 artifact-trust	0.678	0.015	0.582 ($p = 0.060$)

Seal is **essentially uncorrelated** with both structural instruments, which correlate strongly with **each other**. The single point of three-way agreement is **ard25** — a structurally broken ARA (empty trace, missing `logic/experiments.md`) that every instrument independently ranks last. Remove it, and the Seal-vs-structural correlations trend **negative**, while GRO and v3 stay positive with each other. The two mechanical instruments agree about what "complete" looks like; neither agrees with what the calibrated reviewer considers "good."

Illustrative inversions:

- **val25** — Seal's #1 (4.7, Strong Accept) but #6 on GRO health and #10 on v3 trust (0% ref resolvability, `environment_completeness` 0.143). Faithfully scoped, well-evidenced, rigorously falsifiable — with real structural holes.
- **zim25** and **che26** — #1–2 on both structural instruments (GRO 0.986 / 0.982) but only #9–10 on Seal, capped by D6 rigor concerns (sponsor-only authorship, uncorrected multiplicity)

invisible to structural metrics.

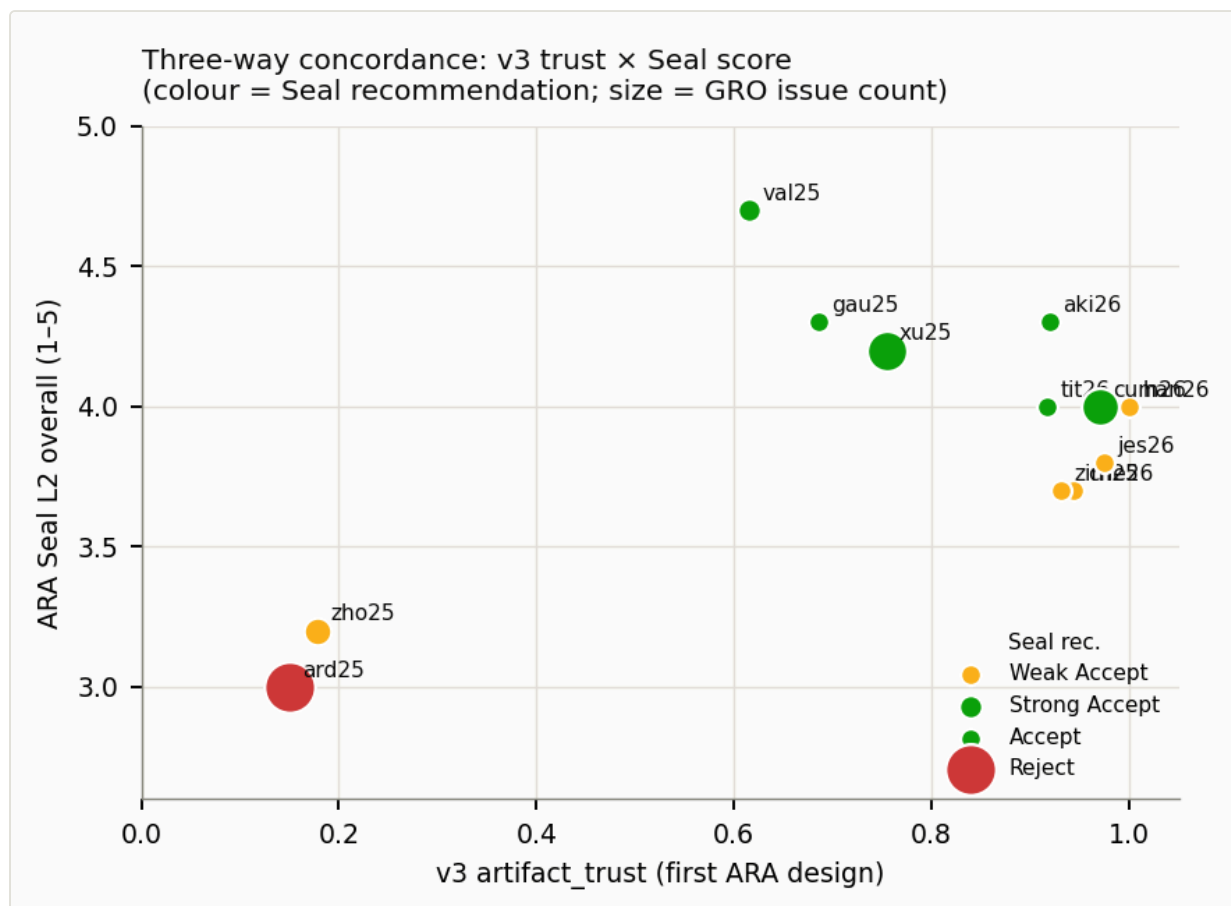


Figure 4. Three-way concordance: v3 artifact_trust $\times$ Seal overall; colour = Seal recommendation, size = GRO issue count.

4.5 The sharpest catch-divergences

Case	v3 (first ARA design)	GRO ideal metric	ARA Seal L2	What it exposes
che26 forest-plot stat	extractive_fidelity 0.625 — catches a fabricated statistic by reading the source	quantity_reconciliation 1.00 — blind (ledger-internal, never reads source)	D6 rigor = 3	determinism is ledger-internal; source fabrication needs the anchored/judged tier
cum26	extractive_fidelity 1.00 (v3's best)	overclaim_flags 9/10 (GRO's worst)	4.0 Accept	faithful extraction ≠ calibrated scope — different axes
xu25	artifact_trust 0.75	overclaim_flags 11/13	4.2 Accept	the mechanical over-claim flag fires on a genre feature Seal judges benign
che26 Co5	—	overclaim_flags = 0 (misses it)	D3 scope = 3, flags over-generalization verbatim	Seal catches scope miscalibration the deterministic flag cannot

ARA	Instrument that catches it	What it catches	The other two
che26	v3 only	Fabricated forest-plot statistic (MD 0.10, CI 0.04–0.16)	GRO 1.0/1.0 (blind); Seal flags a <i>different</i> issue (Co5 scope)
che26	Seal only (D3)	Cross-ethnic overgeneralization from 2 cohorts, no interaction test	GRO overclaim = 0 (n is stated); v3 silent
val25	GRO only	0/55 references resolvable	Seal D1 = 5, overall 4.7; v3 pending
cum26/xu25	GRO flags, arguably wrongly	9/10 and 11/13 "overclaims"	Seal D3 = 4 (genre-appropriate); v3 clean
zim25	Seal only (D6)	Sponsor-only authorship, no code, missing supplements	GRO 0.986 (#1); v3 0.931
ard25	All three	Missing experiments ledger → dangling proof refs	Three-way agreement (the lone case)

5. Synthesis: three complementary tiers

The correlation structure is not a defect in any instrument — it is the empirical signature of the tier gap.

- **Tier A (deterministic — GRO structural, v3 structural components)** answers pure lookups over typed fields: does this ref resolve, does this claim carry a logical form, does this quantity match its quote. Zero comprehension of scientific quality. It catches **catastrophic breakage** (ard25) that no reviewer should read past, and it surfaces **facts a semantic rubric never asks about** (broken-ref counts, 0%-resolvable bibliographies). It is **necessary** — but it correlates near-zero with quality.
- **Tier B (reliable-anchored — GRO retractions / external ref-resolution / registry)** answers "do the external referents exist," via authorities the author cannot edit. It is the tier that **falsified GRO's own deterministic reference_resolvability_rate** (declared 0.52 vs. observed 0.96). Its power in *this* corpus is capped by a floor effect: no known-bad papers, so it is proven trustworthy, not proven discriminating.
- **Tier C (judged / semantic — GRO novelty verdict, Seal L2)** reads the argument and asks whether claims overreach their evidence and whether the methodology is rigorous for the genre. It catches scope overreach, multiplicity risk, and conflict-of-interest that **no structural field encodes**. It is the only tier that tracks "good science" — and it is the most expensive and the least reproducible.

No single approach discriminates good-vs-bad alone. v3's `extractive_fidelity` caught a fabricated statistic that GRO structurally cannot see (che26). GRO caught a 0%-resolvable bibliography that Seal's rubric never examines (val25). Seal caught a cross-ethnic overgeneralization and a sponsor-COI that neither metric can represent (che26, zim25). The three are near-orthogonal signals, and an ARA pipeline needs **Tier A/B structural gates before Tier C semantic review** — a deterministic floor to reject the structurally broken cheaply, an anchored middle to confirm external referents, and a judged ceiling for the epistemic call — **not one instead of the others**.

6. Limitations

We are deliberately conservative about what this comparison establishes.

1. **No external good-vs-bad ground-truth set.** The corpus contains **zero known-bad papers** — 0 retractions, no concealed outcome-switching, all references ultimately real. High reliability on retractions and external ref-resolution reflects a **floor/ceiling effect on a uniformly legitimate corpus**, not proof any instrument would correctly flag a fabricated or retracted paper. GRO's external validation says so directly: this "validates external computability and surfaces concordance/compilation gaps, but it is NOT a full good-vs-bad discrimination test." 2. **The anchored/judged tiers exist for only half the corpus.** All Tier-B/Tier-C metric evidence comes from the 6-ARA p-tau217/donanemab cluster. Every "most uncertain" verdict about anchored/judged metrics is itself demonstrated on **half the corpus, in one narrow clinical/biomarker sub-domain**. The trial-registry concordance finding rests on a **single trial** (NCT04437511) analyzed by two papers. 3. **LLM-authored extensions carry their own uncertainty.** Seal L2 is an LLM semantic reviewer; GRO's novelty verdicts are LLM judgments over PubMed searches whose outcomes are **phrasing-dependent** (22.6% cannot_determine); overclaim_flags is scored over LLM-typed fields. None of these has an independent human adjudicator in this study, so the Tier-C signal is itself unvalidated against expert consensus. 4. **Small, single-domain n.** Twelve ARAs, all Alzheimer's / neurodegeneration, split 6/6 across two clusters. The Spearman correlations ($\rho = -0.011, 0.035, 0.678$) are estimated on $n = 12$ ($n = 11$ excluding ard25); confidence intervals are wide and the near-zero Seal-vs-structural result should be read as "no detectable positive relationship in this sample," not "proven independence." 5. **Known metric defects remain unfixed in the reported numbers.** genre_silent_omissions can go negative (zh025 = -1); broken_ref_integrity conflates missing-ledger with broken-citation (ard25); the deterministic reference_resolvability_rate measures DOI-capture, not verifiability. These are reported as-is for honesty, not corrected.

The defensible conclusion is narrow and, we think, robust: **structural completeness is necessary but not sufficient, the three tiers measure near-orthogonal things on this corpus, and a credible ARA-evaluation pipeline must run all three rather than treat any one as a proxy for the others.**